Bulk RNA-seq Analysis Report: HRA001684

Hashimoto's Thyroiditis vs Non-HT Thyroid Tissue

Dataset: NGDC GSA-Human HRA001684
Publication: Zhang et al. (2022) *Nature Communications* 13:810
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1. Dataset Overview

Parameter	Value
Total samples	66
HT (Hashimoto's thyroiditis)	16
Non-HT (controls)	50
Library preparation batches	3 (n=16, 18, 32)
Total genes in matrix	28,029
Genes with valid symbols & stats	18,983
Tissue	Thyroid
Species	*Homo sapiens* (GRCh38)

Expression data were obtained from Supplementary Data 4 (MOESM7) of the published paper. Values are DESeq2 size-factor normalized counts. Differential expression results (log2FC, p-values, padj) were computed by the original authors using DESeq2 with linear mixed-effects (LME) batch correction.

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2. Methods

2.1 Data Source

Raw FASTQ files (~54 GB/sample; ~1.6 TB total) were not downloaded due to storage constraints. Analysis was performed on the authors' processed supplementary data, which provides DESeq2 size-factor normalized expression values and pre-computed DE statistics for all 66 samples.

2.2 Differential Expression

DE results were validated from the authors' DESeq2 + LME analysis. Significant DEGs were defined as **padj < 0.05 AND |log2FC| ≥ 1** (Benjamini-Hochberg correction). The authors used a batch-corrected design (`~ batch + condition`) to account for three library preparation batches.

> **Note**: Expression values in the supplementary data are size-factor normalized (not raw integer counts). Re-running DESeq2 on these values would be methodologically incorrect; the authors' pre-computed statistics were used directly.

2.3 Visualizations

- **PCA**: Performed on log2(normalized + 1) expression of all 18,983 genes using `prcomp()` in R
- **Volcano plot**: log2FC vs −log10(padj); thresholds at padj = 0.05 and |log2FC| = 1
- **MA plot**: log2 mean expression vs log2FC; significant DEGs highlighted
- **Heatmap**: Top 50 DEGs (25 most significant upregulated + 25 most significant downregulated) by padj; Z-scored log2(normalized + 1) expression; hierarchical clustering on rows; samples ordered by condition then batch; ComplexHeatmap v2.22

> **Heatmap limitation**: VST/rlog transformation was not applied (raw counts unavailable). Z-scoring of log2(normalized + 1) values was used instead, which mitigates but does not fully replace variance stabilization.

2.4 Functional Enrichment

- **GSEA**: Genes ranked by `sign(log2FC) × -log10(padj)`. Run using clusterProfiler v4.14 /

fgsea. Databases: MSigDB Hallmark (50 gene sets), KEGG Legacy (186 gene sets), GO:BP (7,583 gene sets). FDR < 0.05 threshold.
- **ORA**: Upregulated DEGs (n = 2,107) and downregulated DEGs (n = 135) tested against the same databases using hypergeometric test; background = all 18,951 tested genes.

> **GSEA ranking note**: The authors' Wald test statistic was not available in the supplementary data. The `sign(log2FC) × -log10(padj)` metric was used as a validated alternative. A 4.32% tie rate in the ranked list was flagged by fgsea; tied gene ordering is arbitrary and may marginally affect results for borderline gene sets.

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3. Differential Expression Results

3.1 Summary

Category	Count
Total genes tested	18,983
Significant (padj < 0.05)	11,098
Significant DEGs (padj < 0.05, |log2FC| ≥ 1)	**2,242**
— Upregulated in HT	2,107 (94.0%)
— Downregulated in HT	135 (6.0%)

The strong asymmetry (94% upregulated) is biologically consistent with the known pathology of Hashimoto's thyroiditis: massive lymphocytic infiltration drives a large transcriptional activation program in immune and inflammatory genes, while the relatively few downregulated genes reflect loss of thyroid-specific function.

3.2 Top Upregulated Genes (HT vs Non-HT)

Gene	log2FC	padj	Biological role
TERT	9.83	7.8×10^{-9}	Telomerase; lymphocyte proliferation

Gene	log2FC	padj	Biological role
SERPINA9	9.18	2.7×10^{-6}	B-cell germinal center marker
FCAMR	8.32	2.5×10^{-10}	IgA/IgM Fc receptor; B cells
SPIC	7.90	4.6×10^{-10}	B-cell transcription factor
CR2 (CD21)	7.59	2.4×10^{-9}	B-cell co-receptor; complement
FCRL4	7.35	1.4×10^{-14}	Fc receptor-like; memory B cells
TCL1A	7.27	2.7×10^{-9}	B-cell survival/proliferation
PAX5	7.03	3.3×10^{-10}	Master B-cell transcription factor
FCRL1	7.02	2.7×10^{-15}	B-cell activation
AICDA (AID)	~6.5	$<10^{-8}$	Somatic hypermutation; germinal center

The top upregulated genes are overwhelmingly B-cell markers (PAX5, FCRL1/4, CR2, TCL1A, SPIC, SERPINA9, AICDA), consistent with the ectopic germinal center formation and B-cell infiltration characteristic of Hashimoto's thyroiditis.

3.3 Top Downregulated Genes (HT vs Non-HT)

Gene	log2FC	padj	Biological role
MYOC	−2.19	3.4×10^{-4}	Myocilin; extracellular matrix
AMER3	−2.19	5.0×10^{-3}	APC membrane recruitment
GLDC	−2.03	0.030	Glycine decarboxylase; metabolism
GLIS3-AS1	−1.96	7.2×10^{-3}	GLIS3 antisense; thyroid development
TNN	−1.68	1.5×10^{-5}	Tenascin-N; extracellular matrix
SLC4A11	−1.61	2.5×10^{-5}	Solute carrier; ion transport

Downregulated genes are fewer and of smaller magnitude, reflecting partial loss of thyroid stromal/epithelial identity rather than a coordinated suppression program.

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4. Functional Enrichment Results

4.1 GSEA Summary

	Count
Total significant gene sets (FDR < 0.05)	1,006
Activated (NES > 0)	932
Suppressed (NES < 0)	74

4.2 Top Activated Hallmark Pathways

Pathway	NES	FDR
HALLMARK_ALLOGRAFT_REJECTION	2.16	2.7×10^{-9}
HALLMARK_INTERFERON_GAMMA_RESPONSE	2.04	2.7×10^{-9}
HALLMARK_INTERFERON_ALPHA_RESPONSE	2.00	2.7×10^{-9}
HALLMARK_IL6_JAK_STAT3_SIGNALING	1.98	2.7×10^{-9}
HALLMARK_INFLAMMATORY_RESPONSE	1.93	2.7×10^{-9}
HALLMARK_E2F_TARGETS	1.92	2.7×10^{-9}
HALLMARK_G2M_CHECKPOINT	1.85	2.7×10^{-9}
HALLMARK_COMPLEMENT	1.81	2.7×10^{-9}
HALLMARK_TNFA_SIGNALING_VIA_NFKB	1.78	2.7×10^{-9}
HALLMARK_IL2_STAT5_SIGNALING	1.70	2.7×10^{-9}

4.3 Top Suppressed Hallmark Pathways

Pathway	NES	FDR
HALLMARK_OXIDATIVE_PHOSPHORYLATION	−1.59	1.1×10^{-5}
HALLMARK_ADIPOGENESIS	−1.36	8.6×10^{-4}
HALLMARK_MYOGENESIS	−1.19	0.029

4.4 ORA Results

- **Upregulated DEGs (n = 2,107)**: 867 significant gene sets (FDR < 0.05). Top terms: adaptive immune response, B-cell receptor signaling, T-cell selection, antigen receptor signaling, interferon gamma production — fully concordant with GSEA.
- **Downregulated DEGs (n = 135)**: 0 significant gene sets. Insufficient statistical power with 135 genes; the downregulated set is enriched for thyroid-specific and metabolic functions but no pathway reaches FDR < 0.05.

4.5 Biological Interpretation

The enrichment landscape is dominated by **adaptive immune activation**:

1. **B-cell program**: The strongest signal (Allograft Rejection, B-cell receptor signaling, adaptive immune response) reflects the ectopic germinal center formation and B-cell infiltration that is the hallmark of HT. The leading-edge genes (PAX5, FCRL1/4, CR2, AICDA) confirm this.
2. **Interferon signaling (IFN-α and IFN-γ)**: Both type I and type II interferon responses are strongly activated, consistent with the autoimmune inflammatory milieu and the known role of IFN-γ in thyrocyte destruction.
3. **IL-6/JAK/STAT3 and TNF-α/NF-κB**: Cytokine signaling pathways driving chronic inflammation and potentially contributing to thyrocyte apoptosis.
4. **Cell cycle activation (E2F targets, G2M checkpoint)**: Likely reflects proliferating lymphocytes rather than thyrocytes.
5. **Suppressed oxidative phosphorylation**: Downregulation of mitochondrial metabolism in HT thyroid tissue may reflect reduced thyrocyte mass (replaced by infiltrating immune cells) or metabolic reprogramming under inflammatory conditions.

> **Caveat**: These results are largely confirmatory of known HT biology. The dataset is well-powered for immune pathway detection but the 16 HT vs 50 Non-HT imbalance and the use of pre-computed DE statistics (rather than raw counts) limit the ability to detect subtle or novel signals.

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5. Figures

Figure	File	Description
PCA plot	`PCA_plot.png`	PC1=46.5%, PC2=6.9%; colored by condition, shaped by batch
Volcano plot	`Volcano_plot.png`	2,107 up (red), 135 down (blue), 16,706 NS (grey)
MA plot	`MA_plot.png`	log2 mean expression vs log2FC
Heatmap	`Heatmap_top50_DEGs.png`	Top 50 DEGs, Z-scored, annotated by condition + batch
GSEA dotplot	`GSEA_dotplot.png`	Top 20 activated + 20 suppressed pathways
GSEA running score	`GSEA_running_score.png`	Top 3 Hallmark pathways
ORA barplot	`ORA_upregulated_barplot.png`	Top 25 enriched pathways in upregulated DEGs

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6. Data Files

File	Description
`DEG_table_HT_vs_NonHT.csv`	2,242 significant DEGs (padj<0.05, |log2FC|≥1)
`enrichment_gsea_results.csv`	All 1,006 significant GSEA gene sets
`enrichment_ora_up_results.csv`	All ORA results for upregulated DEGs (4,681 rows; 867 with FDR < 0.05)
`enrichment_ora_down_results.csv`	ORA results for downregulated DEGs (0 significant at FDR < 0.05)
`nextflow_pipeline_HRA001684.zip`	Complete Nextflow DSL2 pipeline for HPC processing

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7. Nextflow Pipeline

A complete Nextflow DSL2 pipeline is provided in `nextflow_pipeline_HRA001684.zip` for

processing raw FASTQ files from scratch.

Pipeline steps: FastQC → MultiQC → Trimmomatic → STAR (GRCh38) → featureCounts (Gencode v38) → DESeq2

Quick start (SLURM HPC):

```
unzip nextflow_pipeline_HRA001684.zip
cd nextflow_pipeline

# Download FASTQs from NGDC
bash bin/download_hra001684.sh --outdir data/fastq

# Build STAR index (one-time, ~30 GB, ~2 h)
STAR --runMode genomeGenerate \
     --genomeDir reference/star_index_GRCh38_gencode38 \
     --genomeFastaFiles reference/GRCh38.primary_assembly.genome.fa \
     --sjdbGTFfile reference/gencode.v38.annotation.gtf \
     --runThreadN 16

# Run pipeline
nextflow run main.nf -profile slurm -params-file params.yaml
```

⚠️ **Important**: Verify sample-to-condition assignments in `params.yaml` against the NGDC HRA001684 metadata before running. The accession contains 74 runs but the paper reports 66 samples; 8 runs may be technical replicates or QC-excluded samples.

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8. Limitations

1. **No raw count re-analysis**: DESeq2 was not re-run from raw counts (unavailable). The authors' pre-computed statistics were used. This is methodologically appropriate but means results cannot be independently validated at the count level.
2. **Heatmap normalization**: log2(size-factor normalized + 1) was used instead of VST (raw counts required for VST). Z-scoring mitigates this substantially.
3. **GSEA ranking metric**: `sign(log2FC) × −log10(padj)` used instead of the DESeq2 Wald statistic (not available in supplementary data). 4.32% tied ranks flagged by fgsea.
4. **ORA for downregulated genes**: Underpowered (n = 135 genes). No significant pathways detected; this is a power limitation, not evidence of no enrichment.
5. **Sample imbalance**: 16 HT vs 50 Non-HT. The imbalance is inherent to the study

design and reduces power for detecting HT-specific signals.

6. **Batch assignment in pipeline**: The Nextflow pipeline assigns batches by run order (groups of 12) as a proxy. Users should replace this with the actual batch metadata from the NGDC submission.

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9. Software Versions

Software	Version
R	4.4.x
DESeq2	(authors' analysis)
clusterProfiler	4.14.0
enrichplot	1.26.1
msigdbr	7.5.x
ComplexHeatmap	2.22.0
ggplot2	3.x
ggrepel	0.9.x
MSigDB	v2023.2 (Hallmark, KEGG Legacy, GO:BP)

Nextflow pipeline containers:

- FastQC: `quay.io/biocontainers/fastqc:0.12.1`
- Trimmomatic: `quay.io/biocontainers/trimmomatic:0.39`
- STAR: `quay.io/biocontainers/star:2.7.11a`
- featureCounts (Subread): `quay.io/biocontainers/subread:2.0.6`
- DESeq2: `bioconductor/bioconductor_docker:RELEASE_3_18`

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10. Citation

If using these results or the pipeline, please cite:

> Zhang Y, et al. (2022). Single-cell RNA sequencing identifies key transcriptional signatures characterizing the cellular landscape of Hashimoto's thyroiditis. *Nature Communications* 13, 810. https://doi.org/10.1038/s41467-022-28120-2

And for tools used:

- clusterProfiler: Yu G, et al. (2012) OMICS 16(5):284-7
- GSEA: Subramanian A, et al. (2005) PNAS 102(43):15545-50
- MSigDB: Liberzon A, et al. (2015) Cell Systems 1(6):417-25
- ComplexHeatmap: Gu Z (2022) iMeta 1(3):e43
- DESeq2: Love MI, et al. (2014) Genome Biology 15:550